

WCC SERIES (40 TO 150 TR)

Closed Circuit Evaporative Coil Cooling Towers

Indirect Evaporative Cooling Isolating Process Fluid Inside Sealed Serpentine Heat Exchange Coils for 100% Contamination-Free Cooling

**100% SEALED LOOP**

Process Fluid Never
Contacts Air or Dust

40-150 TR

Packaged Heavy-Duty
Industrial Range

HDG / SS304 COIL

Heavy-Wall Continuous
Serpentine Tubing

ZERO SCALING

Protects Expensive High-
Precision Heat
Exchangers

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WCC Series closed circuit cooling towers provide indirect evaporative cooling for critical applications where process fluid cannot tolerate atmospheric air contamination, dust fouling, biological growth, or mineral scaling. By circulating the process fluid inside a completely sealed, continuous serpentine tube coil while an external deluge water spray and fan provide evaporative cooling, WCC towers safeguard expensive induction furnaces, laser resonators, and vacuum furnaces from catastrophic scale fouling.

Working Principle: Process fluid (clean water, deionized water, or glycol) circulates entirely inside a continuous multi-circuit serpentine coil bundle. Simultaneously, a secondary recirculating pump sprays deluge cooling water over the exterior surface of the coil while top-mounted axial fans draw fresh air upward over the falling water film. A fraction of the spray water evaporates into the air stream, absorbing latent heat and cooling the internal process fluid. The un-evaporated spray water collects in the lower basin and is recirculated. Process fluid remains 100% sealed, pure, and pressurized.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Hot Process Fluid (45°C) → Sealed Serpentine Coil Inlet → Indirect Evaporative Cooling via Deluge Spray + Induced Draft Air → Cooled Process Fluid (32°C) → Clean Factory Return Loop

Completely Sealed Contamination-Free Loop

Eliminates scaling, airborne dust accumulation, bacterial slime, and oxygen ingestion inside client equipment. Eliminates routine chemical descaling of furnace coils, mold channels, and compressor heat exchangers.

Continuous Serpentine Tube Bundle Architecture

Coils are manufactured from continuous smooth-bend heavy-gauge steel tubing, hot-dip galvanized after fabrication in molten zinc (ASTM A123), or 100% SUS304/SUS316L stainless steel for deionized water loops.

Low Water Consumption & Minimal Drift

Recirculates spray water locally, consuming only the water required for evaporation (approx. 1.8 L/hr per kW rejected). High-efficiency drift eliminators limit water carryover to less than 0.005%.

Dual Low-Noise Fans & Non-Clog Deluge Spray

Features dual independent aero-foil fans driven by weather-proof IP55 motors for operational redundancy, coupled with large-orifice clog-free spray nozzles providing uniform coil surface wetting.



CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Process Fluid In: 37°C | Process Fluid Out: 32°C | Wet Bulb Temp: 28°C | Coil Test Pressure: 25 bar g (Air under water) | Casing: Heavy FRP / HDG Panels | Spray Circuit: Low-Head Centrifugal Pump

MODEL CODE	CAPACITY TR	FAN MOTORS	FAN RPM	IN/OUT CONN	COIL VOL LITERS	DIMENSIONS (L×W×H MM)	WEIGHT KG
WCC 40	40 TR	3.0 HP × 2	960 RPM	3" / 3"	72 L	3300 × 1500 × 1400	1,850 kg
WCC 50	50 TR	3.0 HP × 2	960 RPM	3" / 3"	90 L	3300 × 1700 × 1400	2,150 kg
WCC 75	75 TR	3.0 HP × 2	960 RPM	4" / 4"	135 L	3300 × 2000 × 1400	2,850 kg
WCC 100	100 TR	5.0 HP × 2	960 RPM	4" / 4"	180 L	3600 × 2400 × 1600	3,750 kg
WCC 150	150 TR	7.5 HP × 2	720 RPM	6" / 6"	270 L	4200 × 2800 × 1800	5,200 kg

ENGINEERING SELECTION & OPERATIONAL NOTES

- SUS304 or SUS316L stainless steel coil options strictly recommended for deionized water laser loops to prevent zinc leaching.
- Dual fan motors allow 50% power reduction during low-ambient nighttime operation via 2-stage thermostatic switching.
- Integrated winter bypass / dry-cooling operation possible when ambient wet-bulb temperature drops below 10°C.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

- Induction Melting & Billet Heating Furnaces** Protects copper power coils and IGBT power supplies from mineral scale calcification and explosive coil rupture.
- Continuous Casting & Vacuum Furnaces** Delivers scale-free cooling water to vacuum chamber double-walls and turbomolecular pump bearing jackets.
- Industrial Air Compressor Jackets** Replaces open tower cooling on oil-free screw compressor jackets, eliminating cylinder scale and head gasket blowouts.
- Fiber Laser Resonators & Optics** Circulates clean demineralized water in a closed loop, maintaining laser beam alignment without optic fouling.
- Automotive Engine Test Dyno Beds** Absorbs dynamometer engine jacket and exhaust brake heat loads while maintaining closed loop fluid integrity.

Engineering Comparison: Win WCC vs. Open Cooling Tower + Plate Standard ISO 9001:2015 Heat Exchanger Quality		
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
System Footprint	Single monoblock skid integrating coil, spray, basin, and fans	Requires open tower + separate PHE + 2 pump sets + buffer tank
Maintenance	Smooth coil exterior resists scaling; zero internal fouling	Plate heat exchanger clogs with algae every 2–4 months
Pumping Power	Only process pump + small low-head spray pump needed	Dual high-head pump systems consume 35% more electricity
Corrosion Protection	Heavy HDG after coil bending (ASTM A123) or SUS304	Exchanger plates vulnerable to pitting and gasket blowout

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- Continuous pneumatic pressure hold test at 25 bar g submerged in water inspection tank
- Full zinc coating thickness verification (>85 microns) compliant with ASTM A123
- Uniform deluge spray coverage validation across 100% of exterior coil tube surface
- Fan dynamic balance testing and motor electrical insulation breakdown test

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 24 Months warranty on serpentine tube coil bundle; 12 months comprehensive on unit. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

📍 SF No: 4, 195 B, Kallangadu, Arasur Post, Coimbatore – 641407, Tamil Nadu, India

☎ +91 95972 28969 / +91 95972 28975

✉ info@winequipments.com • winequipments.com



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